

Dhruv Talnewar

AI Engineer | AI Product Architect | Agentic & Multi-Agent Systems | Forward Deployed Engineer

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SUMMARY

Results-driven AI Engineer, AI Architect with production-grade experience designing, implementing, and evaluating AI systems across multimodal domains. Proven track record architecting multi-agent LLM pipelines, scalable ML pipelines, and responsible-AI governance frameworks. Fluent in translating research hypotheses into auditable, measurable, and enterprise-deployable AI outcomes. Passionate about core problem-solving, innovations, and the future of intelligent systems.

EXPERIENCE

AI Product Consultant & Freelance AI Solutions Developer

Enterprise & B2B Clients · Ongoing

- Delivered production AI prototypes for enterprise and B2B clients, spanning PRDs, responsible-AI risk frameworks, and scalable ML architecture across autonomous agentic workflows, RAG-based knowledge retrieval, voice-AI agents, 3D and interactive web experiences (React Three Fiber), and workflow automation.

Building Cost-efficient Voice AI Infra.

Voice AI infrastructure layer

- Building a cost-optimized Voice AI infrastructure stack that targets reducing real-time conversational voice costs from industry-standard levels to ₹1–2/min by optimizing the STT, LLM, TTS and telephony pipeline.

ReviewSpark - The Startup Community

- Architected an AI-native feedback infrastructure that converts high-friction customer interactions into structured voice intelligence, retention signals, integrating real-time speech processing, LLM orchestration, privacy-preserving identity, and merchant conversion analytics into a unified multi-tenant system.

PROJECTS

CHD Predictor AI

Python · Scikit-learn · Multimodal Clinical AI

- An agentic, multimodal diagnostic engine that synthesizes high-dimensional clinical data into probabilistic cardiovascular risk profiles and automated, evidence-based intervention protocols, built on feature engineering, k-fold cross-validation, and SHAP-based interpretability for auditable clinical risk scoring. Live: chdpredictor.netlify.app.

EpiSentinel AI

Autonomous Multi-Agent Epidemic Intelligence

- Voice-first epidemic intelligence dashboard providing real-time SEIR forecasting and autonomous alerting across 50 Indian districts, reducing IDSP detection lag from 14 days to under 30 seconds. Live: [EpiSentinel AI](https://episentinel.ai)

Drishti x Titan

Nirman 2026 (Amity) · Top 5, National Finalist

- 8-agent adversarial ML decision council on LangGraph evaluating 19 live financial data streams in real time behind SEBI-compliant risk firewalls; shipped NexusOracle (1,000+ Monte Carlo scenarios per cycle), GeoPulse, and a 38-indicator candlestick pattern recognizer within a 48-hour sprint. Live: [Drishti](https://drishti.ai)

Hackathon MVPs

- CameraAI** (Computer Vision): AI intelligence layer for CCTV monitoring.
- Sentinelx** : Autonomous SOC platform on a 5-agent LangGraph pipeline, with MITRE ATT&CK-mapped detection and CVSS 4.0 risk scoring.
- CogniFlow** (AMD Slingshot 2026, Regional Finalist, Pune): On-device inference under 50ms via AMD Ryzen AI NPU with zero data exfiltration.
- GlitchFree** (Bharatiya Antariksh Hackathon 2026): Air-gapped predictive NOC copilot for ISRO's MPLS network operations.
- Certus Health AI** (Forge the Future Hackathon 2026): Multi-agent healthcare claims and document-intelligence platform.
- Cuexis** (AI Boomi Startup Hack): Responsible-AI governance and prompt-auditing platform; real-time scoring across 4 safety axes with pgvector-backed prompt version control.

TECHNICAL SKILLS

LLM & GenAI Engineering: Multi-provider LLM APIs (Anthropic Claude, Google Gemini, OpenAI GPT, AWS Nova Sonic), prompt engineering & optimization, structured output validation (Zod), function calling & tool use, multimodal inference (vision, voice)

Agentic AI & Orchestration: LangGraph (multi-agent state machines, up to 8-agent pipelines), LangChain, multi-step task decomposition, tool-use design, agent evaluation & model-behavior auditing

RAG & Retrieval Infrastructure: Vector databases (Supabase pgvector, ChromaDB), semantic search, embedding-based retrieval, knowledge-grounded generation

Responsible AI, Safety & Compliance: AWS Guardrails, real-time PII redaction, regulatory-aligned risk firewalls (SEBI, MeitY), safety-axis scoring, post-quantum cryptography (NIST FIPS 203/204/205), MITRE ATT&CK, CVSS 4.0

ML & Applied Research: PyTorch, Scikit-learn, supervised learning, feature engineering, k-fold cross-validation, SHAP interpretability, Monte Carlo simulation

Edge & On-Device AI: Transformers.js, MediaPipe, WebNN, AMD Ryzen AI NPU, on-device SLM inference

Cloud & Deployment: AWS (Bedrock, Fargate, Step Functions), Docker, Firebase/Firestore, Netlify, Pydantic

Data, Backend & Languages: Python, TypeScript, JavaScript, SQL, FastAPI, Node.js, REST, WebSocket, PostgreSQL, Redis, React, Next.js, React Three Fiber, Playwright

ACHIEVEMENTS

- Won the LyzrAI Hackathon - When everyone built AI agents, for everyone I built an Agentic UEBA interface for your multi-agent orchestration.
- Got selected among the top builders for the Junction 2026: European Tech Renaissance.
- National Finalist, AWS AI for Bharat Hackathon 2026. Regional Finalist, AMD Slingshot Hackathon 2026 (Pune), Top 5, Nirman Hackathon 2026 (Amity).
- Won 4 Pune based Hackathons across multiple domains of Healthcare, Fintech,
- Pitched my AI and VR fashion-tech concept startup to VC's and Investors at IIT Bombay for the E-Summit 2025.
- Certifications** - Oracle Agentic AI Certified Foundations Associate, Lyzr Agent Studio and Architect, AWS Fundamentals of ML & Advanced Artificial Intelligence, Google Startup School.

EDUCATION

(B.Tech) - CSE(AI/ML) PES MCOE, Pune | 2024 - 2028

- Projectsspanning FinTech ML, GovTech AI, edge-AI systems, healthcare ML, and responsible-AI governance: all shipped in production environments.